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LAB MANUAL

Topic: Lists and Tuples in Python

Subject: Python Programming

Objective

The objective of this lab manual is to help students understand Lists and Tuples in Python. After completing this lab, students will be able to create lists and tuples, access their elements, use common functions, modify lists, understand tuple immutability, and solve basic programming problems using lists and tuples.

Background

Python provides different data structures to store multiple values in a single variable. Two commonly used data structures are Lists and Tuples.

A List is a collection of items that is ordered, changeable, and allows duplicate values. Lists are useful when we need to store data that may change during program execution.

Example:

```
Python  
numbers = [10, 20, 30, 40]
```



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A Tuple is also a collection of items that is ordered and allows duplicate values, but it is not changeable. Once a tuple is created, its values cannot be modified directly. Tuples are useful when we want to store fixed data.

Example:

Python

```
numbers = (10, 20, 30, 40)
```

Difference Between List and Tuple

- List uses square brackets [].
- Tuple uses round brackets () .
- List is mutable, which means its values can be changed.
- Tuple is immutable, which means its values cannot be changed.
- Lists are used when data needs to be updated.
- Tuples are used when data should remain fixed.

Required Software

- Python 3
- VS Code, PyCharm, Jupyter Notebook, or any Python IDE

Basic Syntax of List:

```
fruits = ["apple", "banana", "mango"]
```

Basic Syntax of Tuple:



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```
fruits = ("apple", "banana", "mango")
```

LISTS IN PYTHON

A list is created by placing values inside square brackets [].

Example:

```
Python  
fruits = ["apple", "banana", "mango"]  
  
print(fruits)
```

Output:

```
None  
['apple', 'banana', 'mango']
```

Common List Functions

- **append()**: Adds an item at the end of the list
- **insert()**: Adds an item at a specific position
- **remove()**: Removes a specific item from the list
- **pop()**: Removes an item by index
- **clear()**: Removes all items from the list



- **sort()**: Sorts the list in ascending order
- **reverse()**: Reverses the list order
- **count()**: Counts how many times a value appears
- **index()**: Returns the index position of a value
- **extend()**: Adds elements of one list to another list
- **copy()**: Creates a copy of the list

Example 1: Creating and Accessing List Elements

Code:

Python

```
students = ["Ali", "Sara", "Ahmed", "Zain"]  
  
print("Complete List:", students)  
  
print("First Student:", students[0])  
  
print("Second Student:", students[1])  
  
print("Last Student:", students[-1])
```

Output:

None

```
Complete List: ['Ali', 'Sara', 'Ahmed', 'Zain']
```

```
First Student: Ali
```



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Second Student: Sara

Last Student: Zain

Explanation:

In this example, a list named students is created. List elements are accessed using index numbers. Positive indexing starts from 0, while negative indexing starts from the end.

Example 2: Adding Elements in a List

Code:

Python

```
fruits = ["apple", "banana", "mango"]
```

Python

```
fruits.append("orange")
```

```
fruits.insert(1, "grapes")
```

Python

```
print(fruits)
```

Output:



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None

```
['apple', 'grapes', 'banana', 'mango', "orange"]
```

Explanation:

The `append()` function adds an item at the end of the list. The `insert()` function adds an item at a specific index position.

Example 3: Removing Elements from a List

Code:

Python

```
fruits = ["apple", "banana", "mango", "orange"]
```

Python

```
fruits.remove("banana")
```

```
fruits.pop(1)
```

Python

```
print(fruits)
```

Output:



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None

```
['apple', 'orange']
```

Explanation:

The `remove()` function removes a specific value from the list. The `pop()` function removes an element using its index number.

Example 4: Sorting and Reversing a List

Code:

Python

```
numbers = [40, 10, 30, 20, 50]
```

Python

```
numbers.sort()
```

```
print("Sorted List:", numbers)
```

Python

```
numbers.reverse()
```

```
print("Reversed List:", numbers)
```



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Output:

None

Sorted List: [10, 20, 30, 40, 50]

Reversed List: [50, 40, 30, 20, 10]

Explanation:

The sort() function arranges the list in ascending order. The reverse() function reverses the current order of the list.

Example 5: Finding Sum, Maximum, Minimum, and Length of a List

Code:

Python

```
marks = [75, 80, 90, 65, 85]
```

Python

```
print("Total Marks:", sum(marks))  
print("Highest Marks:", max(marks))  
print("Lowest Marks:", min(marks))  
print("Number of Subjects:", len(marks))
```

Output:



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None

Total Marks: 395

Highest Marks: 90

Lowest Marks: 65

Number of Subjects: 5

Explanation:

The `sum()` function calculates the total of all list values. The `max()` function finds the highest value. The `min()` function finds the lowest value. The `len()` function counts the number of elements.

Example 6: Looping Through a List

Code:

Python

```
languages = ["Python", "Java", "C++", "JavaScript"]
```

Python

```
for language in languages:  
    print(language)
```

Output:



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None

Python

Java

C++

JavaScript

Explanation:

A for loop is used to access each item of the list one by one.

Solved Program 1: Calculate Average Marks Using List

Problem:

Write a Python program to store marks of five subjects in a list and calculate the total and average marks.

Solution:

Python

```
marks = [78, 85, 90, 66, 72]
```

Python

```
total = sum(marks)
```

```
average = total / len(marks)
```



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Python

```
print("Marks:", marks)

print("Total Marks:", total)

print("Average Marks:", average)
```

Output:

None

```
Marks: [78, 85, 90, 66, 72]
```

```
Total Marks: 391
```

```
Average Marks: 78.2
```

Solved Program 2: Find Even Numbers from a List

Problem:

Write a Python program to display only even numbers from a list.

Solution:

Python

```
numbers = [10, 15, 20, 25, 30, 35, 40]
```

```
print("Even Numbers:")
```



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Python

```
for number in numbers:  
  
    if number % 2 == 0:  
  
        print(number)
```

Output:

None

Even Numbers:

10

20

30

40

Solved Program 3: Count Occurrence of a Value in a List

Problem:

Write a Python program to count how many times 20 appears in a list.

Solution:

Python

```
numbers = [10, 20, 30, 20, 40, 20, 50]
```



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Python

```
count_value = numbers.count(20)
```

Python

```
print("20 appears", count_value, "times")
```

Output:

None

```
20 appears 3 times
```

Solved Program 4: Copy and Extend a List

Problem:

Write a Python program to copy one list and add another list into it.

Solution:

Python

```
list1 = [1, 2, 3]
```

```
list2 = [4, 5, 6]
```



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Python

```
new_list = list1.copy()  
  
new_list.extend(list2)
```

Python

```
print("First List:", list1)  
  
print("Second List:", list2)  
  
print("New List:", new_list)
```

Output:

None

```
First List: [1, 2, 3]
```

```
Second List: [4, 5, 6]
```

```
New List: [1, 2, 3, 4, 5, 6]
```

TUPLES IN PYTHON

A tuple is created by placing values inside round brackets ().

Example:



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Python

```
fruits = ("apple", "banana", "mango")  
  
print(fruits)
```

Output:

None

```
('apple', 'banana', 'mango')
```

Important Tuple Features

- Tuples are ordered.
- Tuples are immutable.
- Tuples allow duplicate values.
- Tuple elements can be accessed using indexing.
- Tuple values cannot be changed directly.

Common Tuple Functions:

count() Counts how many times a value appears in a tuple

index() Returns the index position of a value

len() Returns the number of elements

max() Returns the highest value

min() Returns the lowest value

sum() Returns the total of numeric values

Example 7: Creating and Accessing Tuple Elements



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Code:

Python

```
student = ("Ali", 20, "Artificial Intelligence")
```

Python

```
print("Name:", student[0])  
  
print("Age:", student[1])  
  
print("Department:", student[2])
```

Output:

None

Name: Ali

Age: 20

Department: Artificial Intelligence

Explanation:

Tuple elements are accessed using index numbers. The first element has index 0.

Example 8: Tuple Slicing

Code:



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Python

```
numbers = (10, 20, 30, 40, 50, 60)
```

Python

```
print(numbers[1:4])
```

```
print(numbers[:3])
```

```
print(numbers[3:])
```

Output:

None

```
(20, 30, 40)
```

```
(10, 20, 30)
```

```
(40, 50, 60)
```

Explanation:

Slicing is used to access a specific range of tuple elements.

Example 9: Using count() and index() in Tuple

Code:



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Python

```
numbers = (10, 20, 30, 20, 40, 20)
```

Python

```
print("Count of 20:", numbers.count(20))  
print("Index of 30:", numbers.index(30))
```

Output:

None

Count of 20: 3

Index of 30: 2

Explanation:

The count() function counts how many times a value appears. The index() function returns the first index position of a value.

Example 10: Tuple Unpacking

Code:

Python

```
student = ("Sara", 21, "Computer Science")
```



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Python

```
name, age, department = student
```

Python

```
print("Name:", name)

print("Age:", age)

print("Department:", department)
```

Output:

None

Name: Sara

Age: 21

Department: Computer Science

Explanation:

Tuple unpacking means assigning tuple values to separate variables.

Example 11: Looping Through a Tuple

Code:



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Python

```
subjects = ("Python", "Database", "Mathematics", "AI")
```

Python

```
for subject in subjects:  
    print(subject)
```

Output:

None

Python

Database

Mathematics

AI

Explanation:

A for loop can be used to access tuple elements one by one.

Example 12: Updating Tuple Indirectly

Code:



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Python

```
fruits = ("apple", "banana", "mango")
```

Python

```
temp = list(fruits)

temp[1] = "orange"

fruits = tuple(temp)
```

Python

```
print(fruits)
```

Output:

None

```
('apple', 'orange', 'mango')
```

Explanation:

Tuples cannot be changed directly. To update a tuple, we first convert it into a list, make changes, and then convert it back into a tuple.

Solved Program 5: Find Total and Average Using Tuple

Problem:



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Write a Python program to store marks in a tuple and calculate total and average marks.

Solution:

Python

```
marks = (70, 80, 90, 85, 75)
```

Python

```
total = sum(marks)
```

```
average = total / len(marks)
```

Python

```
print("Marks:", marks)
```

```
print("Total Marks:", total)
```

```
print("Average Marks:", average)
```

Output:

None

```
Marks: (70, 80, 90, 85, 75)
```

```
Total Marks: 400
```



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Average Marks: 80.0

Solved Program 6: Check Item in Tuple

Problem:

Write a Python program to check whether "Python" exists in a tuple or not.

Solution:

Python

```
subjects = ("Python", "Java", "C++", "Database")
```

Python

```
if "Python" in subjects:  
    print("Python is available in the tuple")  
else:  
    print("Python is not available in the tuple")
```

Output:

None

Python is available in the tuple



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Solved Program 7: Convert Tuple into List

Problem:

Write a Python program to convert a tuple into a list.

Solution:

Python

```
numbers = (10, 20, 30, 40)
```

Python

```
numbers_list = list(numbers)
```

Python

```
print("Tuple:", numbers)  
  
print("List:", numbers_list)
```

Output:

None

```
Tuple: (10, 20, 30, 40)
```

```
List: [10, 20, 30, 40]
```




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Solved Program 8: Convert List into Tuple

Problem:

Write a Python program to convert a list into a tuple.

Solution:

Python

```
fruits = ["apple", "banana", "mango"]
```

Python

```
fruits_tuple = tuple(fruits)
```

Python

```
print("List:", fruits)  
  
print("Tuple:", fruits_tuple)
```

Output:

None

```
List: ['apple', 'banana', 'mango']  
  
Tuple: ('apple', 'banana', 'mango')
```



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Solved Program 9: Find Largest and Smallest Number in Tuple

Problem:

Write a Python program to find the largest and smallest number from a tuple.

Solution:

Python

```
numbers = (45, 12, 89, 33, 67)
```

Python

```
largest = max(numbers)
```

```
smallest = min(numbers)
```

Python

```
print("Largest Number:", largest)
```

```
print("Smallest Number:", smallest)
```

Output:

None

```
Largest Number: 89
```



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Smallest Number: 12

Solved Program 10: Use List and Tuple Together

Problem:

Write a Python program to store student information in a tuple and marks in a list. Then display the student information and calculate total marks.

Solution:

Python

```
student = ("Ali", "BS Artificial Intelligence", 20)
marks = [75, 80, 85, 90]
```

Python

```
total = sum(marks)
```

Python

```
print("Student Name:", student[0])
print("Program:", student[1])
print("Age:", student[2])
```



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```
print("Marks:", marks)

print("Total Marks:", total)
```

Output:

```
None
Student Name: Ali
Program: BS Artificial Intelligence
Age: 20
Marks: [75, 80, 85, 90]
Total Marks: 330
```

Practice Tasks

Task 1:

Create a list of five city names. Print the complete list, the first city, and the last city.

Task 2:

Create a list of numbers and display only the odd numbers using a for loop.

Task 3:

Create a list of five student marks. Find the total, average, highest marks, and lowest marks.



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Task 4:

Create a list of fruits. Add two new fruits, remove one fruit, and print the final list.

Task 5:

Create a tuple of five programming languages. Print each language using a for loop.

Task 6:

Create a tuple of numbers. Find the sum, maximum value, minimum value, and length of the tuple.

Task 7:

Create a tuple with repeated values. Use count() to find how many times a specific value appears.

Task 8:

Create a tuple of student information such as name, age, department, and semester. Use tuple unpacking to store each value in a separate variable and print them.

Task 9:

Create a tuple of fruits. Convert it into a list, change one fruit, and convert it back into a tuple.

Task 10:

Create one list and one tuple. Print their types using the type() function and explain the difference between them in comments.

Conclusion

In this lab, we learned about Lists and Tuples in Python. Lists are mutable and can be changed after creation, while tuples are immutable and cannot be changed directly. Lists are useful for dynamic data, whereas tuples are useful for fixed data. We also practiced important functions such as append(), insert(), remove(), pop(), sort(), reverse(), count(), index(), len(), sum(), max(), and min().